Technische Hochschule Deggendorf

Case study Assistive Robotics for improvement of Life Quality

Bibliography Report

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Abstract

The rapid advancement of robotics and communication technologies has enabled the development of telepresence systems that bridge physical distances between individuals. Telepresence robots are increasingly used to support elderly care by allowing remote interaction through video communication and mobility. However, most existing systems rely on manual navigation, limiting usability for elderly users and increasing dependency on remote operators.

This report presents a bibliographic study on telepresence robotic systems and their integration with autonomous navigation technologies. It reviews key advancements in robot navigation, including Simultaneous Localization and Mapping (SLAM), path planning algorithms, and human-robot interaction frameworks. The study highlights the importance of combining telepresence with autonomous mobility to enhance user convenience and system efficiency.

By synthesizing insights from recent research, this report proposes an intelligent telepresence robotic companion capable of executing high-level commands such as “go to kitchen” through autonomous navigation. The proposed system aims to improve elderly care by reducing operational complexity while ensuring safe and efficient movement in indoor environments.

1. Introduction

The increasing aging population has created a demand for assistive technologies that improve the quality of life for elderly individuals. Telepresence robots provide a solution by enabling remote communication between family members and elderly users through video and audio interaction. However, current systems are largely dependent on manual control, requiring continuous user intervention for navigation.

With the rise of autonomous robotics, there is a growing interest in integrating intelligent navigation capabilities into telepresence systems. This integration allows robots to perform tasks such as moving to specific locations without manual control, thereby enhancing usability and independence.

1.1 Objectives

- To review existing literature on telepresence robots and autonomous navigation systems.
- To analyze key technologies such as SLAM, localization, and path planning.
- To identify research gaps in current telepresence robotic systems.
- To propose an autonomous telepresence robotic companion for elderly care.

1.2 Scope of Study

This study focuses on research published between 2018–2025 related to telepresence robotics, indoor navigation, and human-robot interaction. It emphasizes the integration of autonomous navigation with telepresence systems for assistive applications.

2. Bibliography Summary and Literature Alignment

2.1 Overview

This section reviews key research works related to telepresence robots and autonomous navigation. The selected studies focus on robot mobility, environment mapping, obstacle avoidance, and user interaction in indoor environments.

2.2 Summary of Reviewed Research Works

2.2.1 Telepresence Robot with SLAM-Based Autonomous Navigation

This research work focuses on improving telepresence robots by adding autonomous navigation using LiDAR sensors. In this system, a LiDAR sensor is mounted on the robot to scan the surrounding environment. The robot uses a SLAM algorithm called Hector SLAM to create a map of the indoor space while moving.

The system allows the robot to understand its environment and determine its position within that environment. Once the map is created, the robot can navigate to predefined locations such as rooms or specific areas. This reduces the need for continuous manual control by the user. The robot can move independently to target locations based on commands.

However, the system requires good sensor quality and computational power for real-time processing. Also, initial mapping must be done carefully to ensure accuracy.

Key Insight: SLAM enables robots to map and navigate autonomously in real time. **Relevance:** This directly supports the idea of giving commands like “go to kitchen.” **[1]**

2.2.2 Telepresence Robot with Obstacle Avoidance and Navigation Control

This study focuses on improving the safety of telepresence robots by adding obstacle avoidance mechanisms. The robot is designed to detect obstacles such as furniture, walls, or people using sensors. Based on the sensor data, the robot adjusts its movement to avoid collisions.

The system also includes self-localization, which helps the robot understand its position while moving. This is important because the robot must know where it is to

navigate correctly. The study shows that combining localization and obstacle avoidance significantly improves navigation performance.

One limitation is that obstacle avoidance systems must be tuned properly to avoid sudden or unstable movements.

Key Insight: Safe navigation requires real-time obstacle detection and avoidance. **Relevance:** Essential for operating in a home environment with people. **[2]**

2.2.3 Review of Navigation Techniques for Telepresence Robots

This paper provides a detailed review of different navigation methods used in telepresence robots. It discusses three main approaches: manual navigation, semi-autonomous navigation, and fully autonomous navigation.

Manual navigation depends completely on the user, which can be difficult and inefficient. Semi-autonomous systems combine user input with some level of automation. Fully autonomous systems allow robots to move independently without user intervention.

The paper highlights that most current systems are still not fully autonomous and depend heavily on users. It also identifies challenges such as environmental complexity and system integration.

Key Insight: Full autonomy is still not widely implemented in telepresence robots.

Relevance: Shows the need for improved autonomous systems. [3]

2.2.4 ROS-Based Autonomous Navigation System

This research explains how autonomous navigation can be implemented using a robotics framework called ROS (Robot Operating System). The system combines mapping, localization, and path planning into one complete solution.

The robot first builds a map using SLAM. Then, it uses a localization method called AMCL to determine its position on the map. After that, path planning algorithms are used to move the robot to a target location.

The advantage of this system is that it is modular and flexible, allowing easy integration of different components. However, it requires proper setup and parameter tuning.

Key Insight: ROS provides a complete system for autonomous robot navigation. **Relevance:** Useful for implementing real-time navigation in your project. **[4]**

2.2.5 Shared Autonomy in Telepresence Robots

This study introduces a shared control system where both the user and the robot share control over navigation. The robot can automatically avoid obstacles and people, while the user can still guide it when needed.

This approach improves safety and usability because the robot can handle difficult situations automatically. It also reduces the workload on the user.

However, the system is not fully autonomous, as it still depends on user input.

Key Insight: Combining human control and autonomy improves performance. **Relevance:** Can be used as an intermediate step towards full autonomy. **[5]**

2.2.6 SLAM for Autonomous Robots

This study explains how mapping and localization are performed together using the concept of Simultaneous Localization and Mapping . SLAM allows the robot to build a map while also tracking its position.

This is one of the most important technologies in robotics. Without SLAM, the robot cannot navigate unknown environments. The study also discusses different types of SLAM, including LiDAR-based and vision-based approaches. **[6]**

2.2.7 Path Planning Algorithms for Mobile Robots

This research explains how robots calculate the best path from one location to another. Algorithms like A* are used to find the shortest and safest path.

The robot considers obstacles and chooses a path that avoids them while minimizing distance. Path planning is essential for efficient navigation. **[7]**

2.2.8 Obstacle Avoidance Using Dynamic Window Approach

This paper introduces a real-time obstacle avoidance method called the Dynamic Window Approach (DWA). The robot calculates possible movements and selects the safest one based on its speed and surroundings.

This method is useful in dynamic environments where obstacles may move. **[14]**

2.2.9 Hector SLAM for Real-Time Mapping

This paper introduces a real-time obstacle avoidance method called the Dynamic Window Approach (DWA). The robot calculates possible movements and selects the safest one based on its speed and surroundings.

This method is useful in dynamic environments where obstacles may move. **[14]**

2.2.10 AMCL Localization for Mobile Robots

This research explains how robots estimate their position using probability-based methods. AMCL uses multiple particles to represent possible positions and updates them based on sensor data.

This method is widely used in autonomous systems.

Key Insight: Reliable localization improves navigation accuracy.
Relevance: Essential after mapping is completed. [10]

2.3 Summary of Reviewed Works

The reviewed research shows that telepresence robots are widely used for communication but lack autonomous navigation. Technologies such as SLAM, path planning, and obstacle avoidance are well developed but are not fully integrated into telepresence systems.

Most systems focus on either communication or navigation, but not both together. This creates a gap in developing fully autonomous telepresence robots.

Criteria	Research Insight	Proposed Model Justification
Navigation	SLAM Required	SLAM based Mapping
Localization	AMCL Used	Accurate Positioning
Path Planning	A* algorithm	Efficient navigation
Safty	Obstacle avoidance	Real – time safty
Interaction	Voice Control	Easy commands

Table 2.1: Alignment of reviewed research insights with the proposed model selection.

2.4 Synthesis and Observations

From the literature, several important observations can be made:

• Telepresence robots are effective for communication but depend on manual control

• Autonomous navigation technologies are advanced and reliable

• SLAM is essential for mapping and localization

• Path planning and obstacle avoidance improve safety

• Human-robot interaction plays an important role in usability

Overall, there is a clear need to combine these technologies into one system.

3. Model Selection Justification

The development of an autonomous telepresence robotic companion requires careful selection of hardware components and software frameworks. The selected models and technologies must provide a balance between accuracy, real-time performance, cost, and ease of implementation. Based on the literature review, key components such as SLAM algorithms, localization methods, sensors, and software frameworks were chosen to ensure reliable and efficient system performance.

For navigation, SLAM-based mapping was selected because it allows the robot to build a map of the environment while simultaneously tracking its position. This is essential for indoor environments where maps are not pre-defined. Among different SLAM approaches, LiDAR-based SLAM is preferred due to its high accuracy and independence from lighting conditions.

For localization, AMCL (Adaptive Monte Carlo Localization) is selected because it provides reliable position estimation within a known map. It uses probabilistic techniques to handle uncertainty and noise in sensor data, which improves navigation accuracy.

For system integration, ROS (Robot Operating System) is chosen as the main framework. ROS provides pre-built modules for mapping, localization, path planning, and obstacle avoidance. This reduces development time and ensures compatibility between different components.

In terms of interaction, a voice-based command system is selected to make the robot easy to use, especially for elderly users. Voice commands such as “go to kitchen” simplify operation compared to manual control using joysticks or mobile apps.

Overall, the selected models and technologies work together to create a system that is autonomous, safe, and user-friendly.

3.1 Reasons for Selection (Sensors & Components)

Sensor / Component	**Function**
LiDAR Sensor	Measures distances using laser beams and creates accurate maps for navigation.
RGB / RGB-D Camera	Captures video for telepresence and provides visual/depth information.
Wheel Encoders	Measure wheel rotation to estimate robot movement and support localization.

Table 3.1 : Sensors and Components Used in the Proposed Telepresence Robotic System

3.2 Conclusion

The study of telepresence robots and autonomous navigation systems shows that there is a strong need to integrate communication and mobility into a single intelligent system. While current telepresence robots are effective for remote interaction, they rely heavily on manual control, which limits their usability, especially for elderly individuals.

This report analyzed various research works related to SLAM, localization, path planning, obstacle avoidance, and human-robot interaction. The findings indicate that these technologies are well developed individually but are not fully combined in existing telepresence systems.

To address this gap, this study proposes an autonomous telepresence robotic companion capable of executing high-level commands such as “go to kitchen.” By integrating LiDAR-based SLAM, AMCL localization, ROS navigation framework, and voice-based interaction, the system can operate independently in indoor environments.

In conclusion, the integration of autonomous navigation with telepresence robots represents an important step toward intelligent assistive systems. This approach has the potential to significantly improve the quality of life for elderly individuals and contribute to the development of advanced robotic solutions in healthcare and smart home environments.

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